

HR Analytics: Employee Attrition Prediction

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1. Introduction

Employee attrition is costly: hiring, training and lost productivity can cost about half of an employee's annual salary. This project analyses the IBM HR Analytics dataset (1,470 employees, 35 attributes; a fictional practice dataset) to find why employees leave, predict who is at risk, and estimate the value of targeted retention.

2. Abstract

Data was cleaned, new features were engineered (income vs job-level average, promotion-gap ratio), and attrition patterns were tested statistically. Four classifiers were compared using stratified 5-fold cross-validation and the best (Logistic Regression, CV ROC-AUC 0.83) was tuned for high recall. SHAP explained individual and overall drivers. Every employee received a Low / Medium / High risk score, shown in a Power BI dashboard. Overtime, low pay, short tenure and frequent travel were the strongest risk factors.

3. Tools Used

Tool	Purpose
Python (Pandas, NumPy, SciPy)	Cleaning, feature engineering, chi-square and t-tests
Scikit-learn, XGBoost	Logistic Regression, Decision Tree, Random Forest, XGBoost, cross-validation, metrics
SHAP, Seaborn, Matplotlib	Model explainability and visualisation
Power BI (DAX)	Two-page interactive dashboard: Overview and Employee Risk

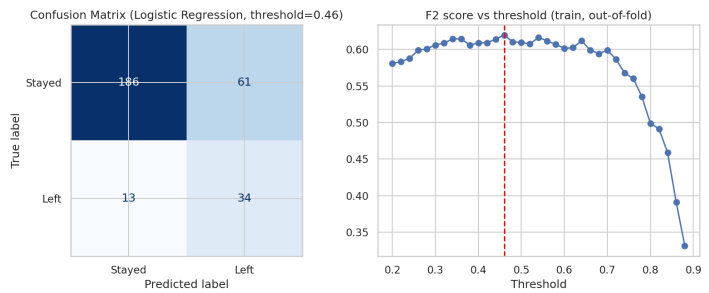
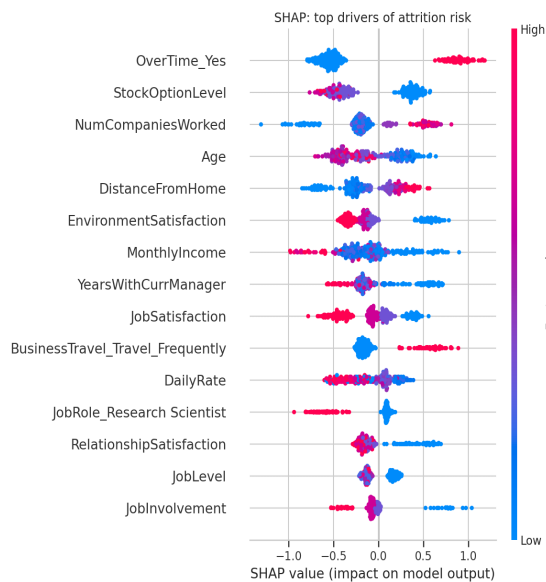
4. Steps Involved in Building the Project

- **Cleaning:** no missing values or duplicates; dropped constant columns (EmployeeCount, Over18, StandardHours) and the ID column. Attrition rate is 16.1%, so the data is imbalanced.
- **EDA and statistical tests:** attrition by department, role, income band, tenure and overtime; chi-square tests (categorical) and Welch t-tests (numeric) confirmed which patterns are significant.
- **Modelling:** stratified 80/20 split, class weights for imbalance, 5-fold CV on training data, one final test on unseen data. Models were compared on Recall, Precision, F1 and ROC-AUC, not accuracy alone.
- **Threshold tuning:** chosen on out-of-fold training predictions by maximising F2 (recall weighted higher), because missing a leaver costs more than a false alarm.
- **SHAP and risk scoring:** SHAP for global and individual explanations; cross-validated risk scores for all employees exported to Power BI.

5. Key Findings

Area	Finding
Overtime	30.5% attrition with overtime vs 10.4% without; #1 SHAP driver
Pay and role	Lowest income band 29.3% vs about 10% in top bands; Sales Representatives 39.8%
Tenure and profile	First 2 years 29.8%; single employees 25.5%; frequent travel and long commute add risk
Not significant	Gender, last salary-hike % and years since last promotion ($p > 0.05$)
Risk buckets	High-risk group (454 employees) had 37.7% actual attrition vs 4.2% in Low

6. Model Results and Explainability



Confusion matrix on 294 unseen employees (threshold 0.46): recall 72%, precision 36%, ROC-AUC 0.81.

Model	CV AUC	Recall	F1
Logistic Reg.*	0.831	0.64	0.45
XGBoost	0.808	0.47	0.44
Random Forest	0.802	0.11	0.16
Decision Tree	0.697	0.53	0.42

*Selected model. Metrics at default 0.5 threshold, test set.

SHAP summary (left): each dot is an employee; right side raises attrition risk. Overtime, few stock options, many past employers, young age and long commute raise risk most.

7. Conclusion and Recommendations

The model flags 72% of leavers in the High-risk bucket (171 of 237). Assuming replacement cost of 50% of annual salary, these leavers cost about \$4.1M; if targeted retention saves 30% of them, the estimated saving is about \$1.2M per year (assumption-based).

- **Overtime:** review workload and overtime policy, especially in Sales and Laboratory Technician roles.
- **Pay and stock options:** review the lowest income band and employees paid below their job-level average.
- **Onboarding:** mentoring and check-ins during the first two years.
- **Stay interviews:** use the High-risk list monthly. Scores support, not replace, manager judgement.

Limitations: fictional dataset; results show association, not causation.